



Advanced SOC Threat Hunting and Incident Response Automation Report

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INTRODUCTION:

Advanced SOC operations involve proactive threat hunting, incident response automation, adversary emulation, evidence analysis, and executive reporting procedures used to improve cybersecurity monitoring capabilities within enterprise environments. This practical task was performed to understand advanced SOC methodologies and investigate how security teams proactively identify threats, automate response workflows, analyze digital evidence, and improve incident handling processes.

Various practical activities including threat hunting, SOAR playbook development, root cause analysis, alert triage, adversary emulation, and security metrics reporting were documented throughout the task. Different monitoring commands, automation concepts, and attack simulation procedures were studied to improve understanding of modern SOC workflows and incident response operations. The capstone project further demonstrated a complete SOC investigation lifecycle including attack detection, containment, analysis, reporting, and continuous improvement procedures.



► Objective:

- ☐ To understand proactive threat hunting methodologies
- ☐ To learn SOAR automation and playbook development
- ☐ To perform root cause analysis and post-incident review
- ☐ To analyze and triage security alerts
- ☐ To study adversary emulation techniques
- ☐ To understand evidence preservation procedures
- ☐ To calculate SOC security metrics and executive reporting
- ☐ To simulate a complete SOC incident response workflow

TOOLS USED:

- Kali Linux
- Elastic Security
- Wazuh
- AlienVault OTX
- VirusTotal
- Metasploit Framework
- Google Sheets
- Draw.io
- GitHub

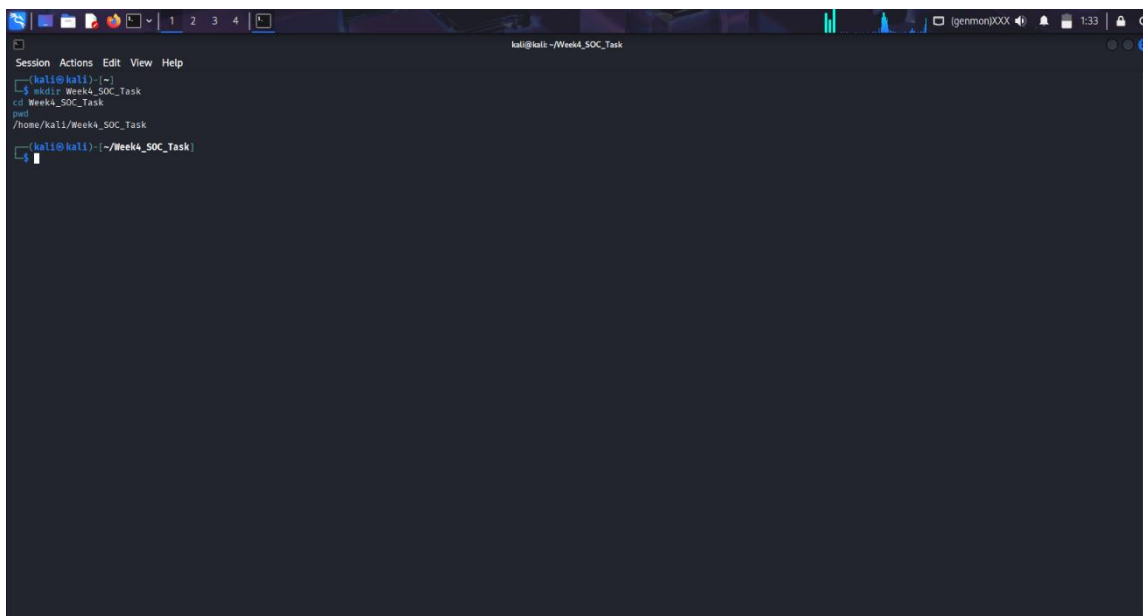


Figure 1: SOC Environment Setup in Kali Linux



► Explanation:

The working directory for the Week 4 SOC practical activities was created and verified using Kali Linux terminal commands. This environment was used to organize screenshots, evidence files, investigation notes, and workflow documentation related to advanced SOC operations and incident response simulations.

► Observation:

The environment setup process successfully created a dedicated workspace for storing SOC investigation artifacts and practical task documentation required during the advanced SOC workflow exercises.

► Learning Outcome:

This activity improved understanding of Linux file management procedures and the importance of maintaining organized working environments during cybersecurity investigations and SOC monitoring operations.

SECTION 1 - Threat Hunting Practice

► Objective:

Threat hunting activities were performed to proactively identify suspicious processes, abnormal network activity, and potential unauthorized access attempts within the monitored environment using system monitoring and log analysis techniques.

The activity also focused on improving visibility into system behavior by examining network traffic patterns and running processes. Threat hunting techniques were used to identify indicators of compromise (IOCs) and enhance the ability to detect potential security incidents before they escalate into major threats.





```
kali@kali: ~$ ss -tulnp
State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port      Process
LISTEN     0           511         127.0.0.1:5601          *:*
LISTEN     0           4096        0.0.0.0:9200            *:*
LISTEN     0           4096        0.0.0.0:9300            *:*
LISTEN     0           4096        0.0.0.0:9300            *:*
```

Figure 4: Active Port and Service Enumeration Using the ss Command

THREAT HUNTING TABLE:

Timestamp	User	Event ID	Notes
2026-05-27 10:00:00	testuser	4672	Unexpected admin role
2026-05-27 10:05:00	admin01	4624	Suspicious login activity
2026-05-27 10:10:00	analyst	4688	PowerShell execution detected

► Explanation:

Threat hunting activities were performed to proactively identify suspicious system behavior and potential attacker activity within the monitored environment. Various monitoring commands were executed to analyze active network connections, running system processes, and listening ports. The collected information was reviewed to identify unusual activities and improve visibility into potential security threats during monitoring operations.



► Observation:

The monitoring activities provided visibility into active system processes, established network sessions, and open listening ports within the environment. The analysis identified multiple active services and system-level processes commonly observed during SOC monitoring operations.

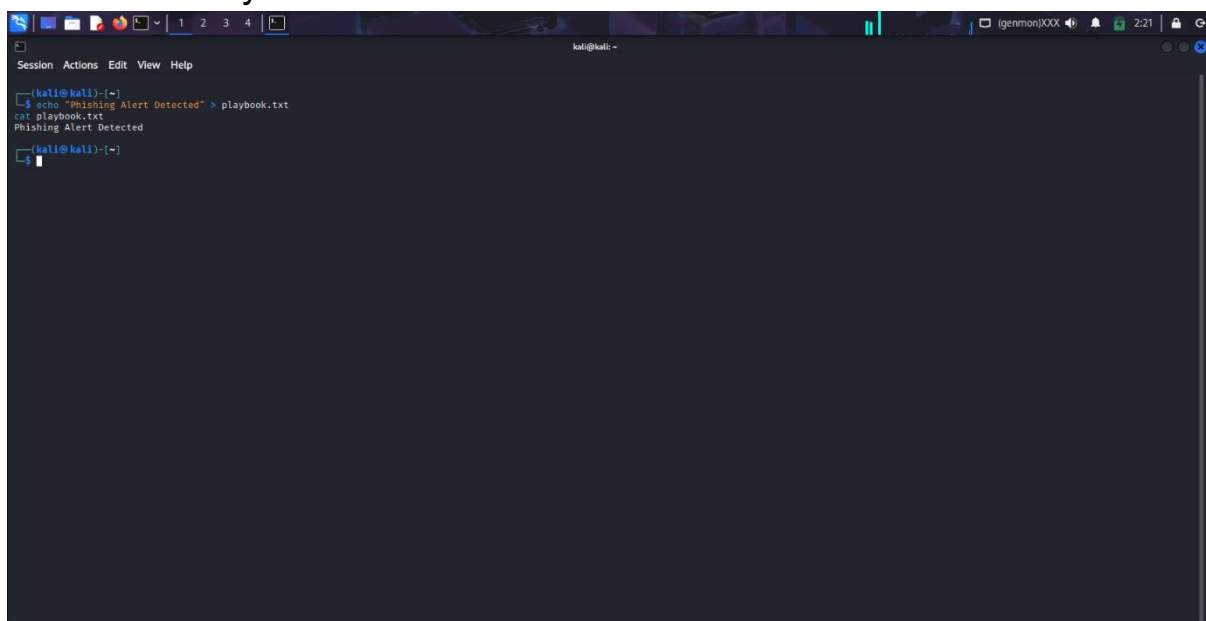
► Learning Outcome:

This activity improved understanding of proactive threat hunting methodologies, network monitoring techniques, process analysis, and suspicious activity identification procedures commonly followed by SOC analysts during threat investigation activities.

Section 2 - SOAR Playbook Development

► Objective:

The objective of this activity was to understand Security Orchestration, Automation, and Response (SOAR) concepts by designing and documenting an automated incident response workflow for phishing-related security alerts.



```
kali@kali:~$ echo "Phishing Alert Detected" > playbook.txt
kali@kali:~$ cat playbook.txt
Phishing Alert Detected
```

Figure 5: Simulated SOAR Playbook Alert Generation



PLAYBOOK WORKFLOW TABLE:

Playbook Step	Status	Notes
Check IP Reputation	Success	IP identified as malicious
Block Suspicious IP	Success	CrowdSec blocked IP
Create TheHive Ticket	Success	Incident ticket generated
Notify SOC Analyst	Success	Email notification triggered
Escalate High Severity Alert	Success	Tier 2 escalation completed

► Explanation:

The SOAR playbook development activity was performed to understand how automated response workflows improve SOC operational efficiency during security incidents. A simulated phishing response workflow was designed to automate IP reputation validation, malicious IP blocking, incident ticket creation, and analyst notification procedures. The workflow demonstrated how automation reduces manual effort and accelerates incident response coordination.

► Observation:

The automation workflow successfully simulated phishing incident response procedures and demonstrated how automated playbooks improve alert handling consistency. Automated escalation and ticket creation reduced manual investigation delays during incident response operations.

► Learning Outcome:

This activity improved understanding of SOAR automation concepts, playbook development procedures, and automated incident response workflows commonly implemented within modern SOC environments.



SECTION 3 - POST-INCIDENT ANALYSIS

► Objective:

The objective of this activity was to perform post-incident analysis and identify the root cause of a simulated phishing incident using structured investigation methodologies and security review procedures.

WHYS TABLE:

Questions	Answers
Why was the phishing email opened?	User clicked malicious link
Why was the link clicked?	Weak email filtering
Why did filtering fail?	Outdated spam rules
Why were rules outdated?	Lack of regular updates
Why were updates delayed?	Insufficient maintenance procedures

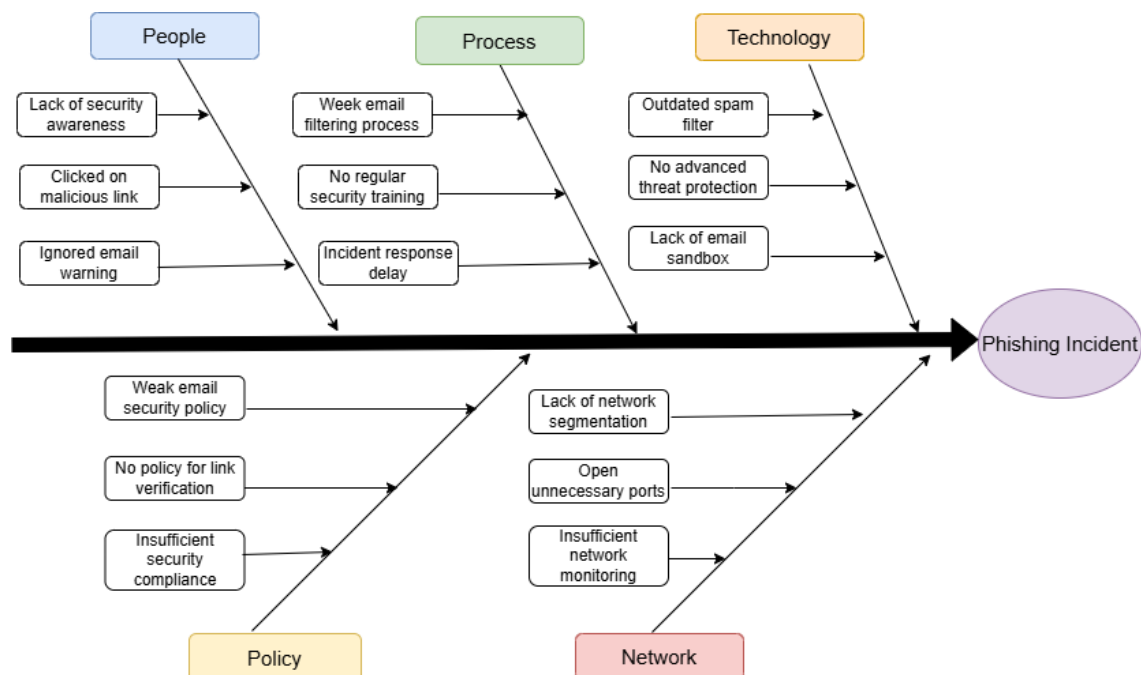


Figure 6: Fishbone Diagram for Root Cause Analysis



► **Explanation:**

Post-incident analysis activities were performed to identify the root cause of the simulated phishing incident and evaluate weaknesses within the existing security controls. The investigation used the 5 Whys methodology and Fishbone Diagram analysis to identify contributing factors related to user awareness, email filtering, system maintenance, and operational procedures. The analysis improved understanding of structured incident review processes commonly followed within SOC environments.

► **Observation:**

The root cause analysis identified weak email filtering mechanisms and insufficient security awareness as major contributing factors during the simulated phishing scenario. The investigation also highlighted the importance of maintaining updated security controls and regular monitoring procedures.

► **Learning Outcome:**

This activity improved understanding of root cause analysis methodologies, post-incident review procedures, and continuous improvement practices commonly followed during SOC investigations and cybersecurity incident response operations.

SECTION 4 - ALERT TRIAGE WITH AUTOMATION

► **Objective:**

The objective of this activity was to analyze and prioritize security alerts using automated validation techniques, threat intelligence concepts, and hash verification procedures commonly followed during SOC investigations. The exercise also aimed to improve decision-making during incident investigations by categorizing alerts according to their risk level and potential impact. Automated validation methods were studied to reduce investigation time and improve the accuracy of alert prioritization procedures.



```
kali@kali: ~  
$ echo "test" > test.txt  
kali@kali: ~  
$ sha256sum test.txt  
f2ca1bb6c7e9b7d06dfe4687e579fce76b37e4e93b7605022da52e6ccc26fd2  test.txt  
kali@kali: ~  
$
```

Figure 7: SHA256 Hash Verification for Alert Validation

ALERT TRIAGE TABLE:

Alert ID	Description	Source IP	Severity	Status
ALT-101	Suspicious File Download	192.168.1.102	High	Investigating
ALT-102	Malware Activity Detected	192.168.1.105	Critical	Escalated
ALT-103	Multiple Failed Logins	192.168.1.110	Medium	Open

IOC VALIDATION TABLE:

IOC Type	Value	Verification Status
SHA256 Hash	f2ca1bb6...	Verified
IP Address	192.168.1.105	Suspicious
Domain	phishing-alert.com	Malicious



► **Explanation:**

Alert triage activities were performed to validate suspicious indicators and prioritize alerts based on severity and potential impact. Hash verification techniques were used to validate evidence integrity and identify potentially malicious files during investigation activities. The collected indicators were reviewed using simulated threat intelligence concepts to improve alert classification accuracy and incident prioritization procedures.

► **Observation:**

The alert triage process successfully demonstrated how automated validation techniques improve investigation accuracy and reduce false positives during SOC operations. Hash verification and indicator analysis improved visibility into suspicious activities and assisted in prioritizing high-severity incidents.

► **Learning Outcome:**

This activity improved understanding of alert prioritization methodologies, indicator validation procedures, hash verification techniques, and automated SOC triage workflows commonly used during cybersecurity investigations.

SECTION 5 - EVIDENCE ANALYSIS

► **Objective:**

The objective of this activity was to collect, preserve, and analyze digital evidence while maintaining proper investigation documentation and evidence integrity procedures during SOC investigation activities. Additional focus was placed on understanding the importance of evidence preservation and maintaining a clear chain of custody throughout the investigation process. Proper evidence handling practices help ensure the reliability and integrity of findings during cybersecurity investigations.



```
Session Actions Edit View Help
(kali@kali)~$ ls -la
total 22368
drwxr-xr-x 28 kali kali 4096 May 28 02:21 .
drwxr-xr-x 3 root root 4096 May 29 2025 ..
-rw-r--r-- 1 kali kali 220 May 29 2025 .bash_logout
-rw-r--r-- 1 kali kali 5551 May 29 2025 .bashrc
-rw-r--r-- 1 kali kali 3526 May 29 2025 .bashrc.original
drwxr-xr-x 8 kali kali 4096 Nov 5 2025 .BurpSuite
drwxr-xr-x 18 kali kali 4096 Apr 27 03:55 .cache
-rw-r--r-- 1 root root 22633970 Apr 27 01:38 capture.pcap
drwxr-xr-x 18 kali kali 4096 May 13 02:28 .config
drwxr-xr-x 3 kali kali 4096 Apr 23 04:13 .dbus
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Desktop
-rw-r--r-- 1 kali kali 35 Sep 6 2025 .dmrc
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Documents
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Downloads
-rw-r--r-- 1 kali kali 13 May 22 01:54 evidence.txt
-rw-r--r-- 1 kali kali 11759 May 29 2025 .face
lrwxrwxrwx 1 kali kali 5 May 29 2025 .face.icon -> .face
drwxr-xr-x 3 kali kali 4096 Sep 6 2025 .gnupg
drwxr-xr-x 2 kali kali 4096 Apr 23 04:13 .gvfs
-rw-r--r-- 1 kali kali 0 Sep 6 2025 .ICEauthority
drwxr-xr-x 4 kali kali 4096 Oct 10 2025 .java
drwxr-xr-x 6 kali kali 4096 Apr 27 01:48 .local
drwxr-xr-x 4 kali kali 4096 Oct 1 2025 .mozilla
drwxr-xr-x 12 kali kali 4096 Apr 27 01:12 .msf4
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Music
-rw-rw-r-- 1 kali kali 20435 Apr 27 02:00 packet_sizes.png
drwxr-xr-x 2 kali kali 12288 May 28 04:22 Pictures
drwxr-xr-x 3 kali kali 4096 Oct 10 2025 .pki
-rw-rw-r-- 1 kali kali 24 May 28 02:21 playbook.txt
drwxr-xr-x 2 kali kali 4096 Oct 15 2025 .praneeth
-rw-r--r-- 1 kali kali 887 May 29 2025 .profile
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Public
drwxr-xr-x 7 kali kali 4096 Oct 16 2025 Storm-Breaker
-rw-r--r-- 1 kali kali 0 Sep 9 2025 sudo_as_admin_successful
-rw-r--r-- 1 root root 112 Apr 21 02:18 synscan.txt
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Templates
-rw-rw-r-- 1 kali kali 209 Apr 24 12:54 test_packets.py
-rw-rw-r-- 1 kali kali 210 Apr 26 12:51 test_packets.pyyyy
-rw-rw-r-- 1 kali kali 5 May 28 04:21 test.txt
-rw-rw-r-- 1 kali kali 0 Apr 27 01:48 To
-rw-rw-r-- 1 kali kali 662 Apr 27 04:01 traffic_analyzer.py
-rw-rw-r-- 1 kali kali 53 Apr 27 01:42 traffic_analyzer.py.save
drwxr-xr-x 2 kali kali 4096 Sep 6 2025 Videos
drwxr-xr-x 2 kali kali 4096 May 13 02:11 Week2_SOC_Practical
```

Figure 8: File and Directory Evidence Analysis

```
Session Actions Edit View Help
(kali@kali)~$ history
1  ls
2  cd Storm-Breaker
3  ls
4  cd
5  sudo python3 st.py
6  ls
7  clear
8  ls
9  cd Storm-Breaker
10 ls
11 sudo python3 st.py
12 clear
13 ls
14 sudo python3 st.py
15 ls
16 sudo python3 st.py
17 exit
18 ls
19 cd Storm-Breaker
20 ls
21 sudo python3 st.py
22 exit
23 sudo systemctl restart networkmanager
24 sudo systemctl restart NetworkManager
25 sudo dhclient
26 exit
27 ls
28 cd Storm-Breaker
29 ls
30 sudo python3 st.py
31 ls
32 sudo python3 st.py
33 exit
34 nmap -v
35 nmap 127.0.0.1
36 nmap 192.168.1.1
37 sudo iptables -f
38 sudo apt update
39 sudo apt install openvas -y
40 sudo apt update
41 sudo apt install openvas -y
42 nano test_packets.py
43 python3 test_packets.py
44 sudo python3 test_packets.py
45 sudo apt update
46 sudo apt install snort -y
```

Figure 9: Command History Investigation



CHAIN OF CUSTODY TABLE:

Evidence Item	Description	Collected By	Date	Integrity Status
Network Log	Active connection records	SOC Analyst	2026-05-22	Verified
Process List	Running process analysis	SOC Analyst	2026-05-22	Verified
Command History	Terminal activity logs	SOC Analyst	2026-05-22	Verified
Test File Hash	SHA256 validation result	SOC Analyst	2026-05-22	Verified

EVIDENCE ANALYSIS TABLE:

Evidence Type	Source	Observation
System processes	ps aux	Multiple active services
Network Session	netstat -an	Established connection detected
File Evidence	ls -la	Test evidence files identified
User Activity	history	Monitoring commands executed

► Explanation:

Evidence analysis activities were performed to review system-level artifacts, terminal activity logs, and collected investigation data during the simulated SOC investigation process. File listings, command history records, and previously generated evidence files were analyzed to understand system activity and maintain proper investigation documentation. Chain-of-custody procedures were followed to ensure evidence integrity and proper handling throughout the investigation process.



MITRE ATT&CK TABLE:

Timestamps	MITRE Technique	Technique ID	Detection Status
2026-05-27 17:00:00	Phishing Simulation	T1566	Detected
2026-05-27 17:10:00	Exploitation Attempt	T1210	Detected
2026-05-27 17:20:00	Command Execution	T1059	Monitored

ATTACK SIMULATION TABLE:

Activity	Tool Used	Result
Attack Simulation	Metasploit	Successful
Network Monitoring	netstat	Connection identified
Process Monitoring	Ps aux	Processes Monitored
Alert Generation	Wazuh	Alert Triggered

► Explanation:

Adversary emulation activities were performed to simulate attacker behavior and evaluate SOC detection capabilities during controlled attack scenarios. The Metasploit Framework was used to simulate exploitation and command execution activities aligned with the MITRE ATT&CK framework. Monitoring procedures were used to observe system behavior, detect suspicious activity, and validate incident response readiness during the simulation process.

► Observation

The adversary emulation exercise successfully demonstrated how simulated attacker activities generate detectable indicators within monitored environments. The activity also highlighted the importance of



continuous monitoring, alert correlation, and proactive threat detection procedures during SOC operations.

► Learning Outcome:

This activity improved understanding of adversary emulation methodologies, MITRE ATT&CK mapping techniques, attacker simulation concepts, and SOC monitoring validation procedures commonly used during cybersecurity defense operations.

SECTION 7 - SECURITY METRICS & EXECUTIVE REPORTING

► Objective:

The objective of this activity was to evaluate SOC operational performance using security metrics and generate executive-level reporting to summarize incident response effectiveness and monitoring improvements.

SOC METRICS TABLE:

Metric	Value
Mean Time to Detect (MTTD)	2 Hours
Mean Time to Respond (MTTR)	4 Hours
Dwell Time	6 Hours
False Positive Rate	8%
Incident Resolution Rate	92%



INCIDENT SUMMARY TABLE:

Incident Type	Count
Phishing Alerts	12
Malware Alerts	8
Failed Login Attempts	15
Suspicious Connections	6
Unauthorized Access Attempts	4

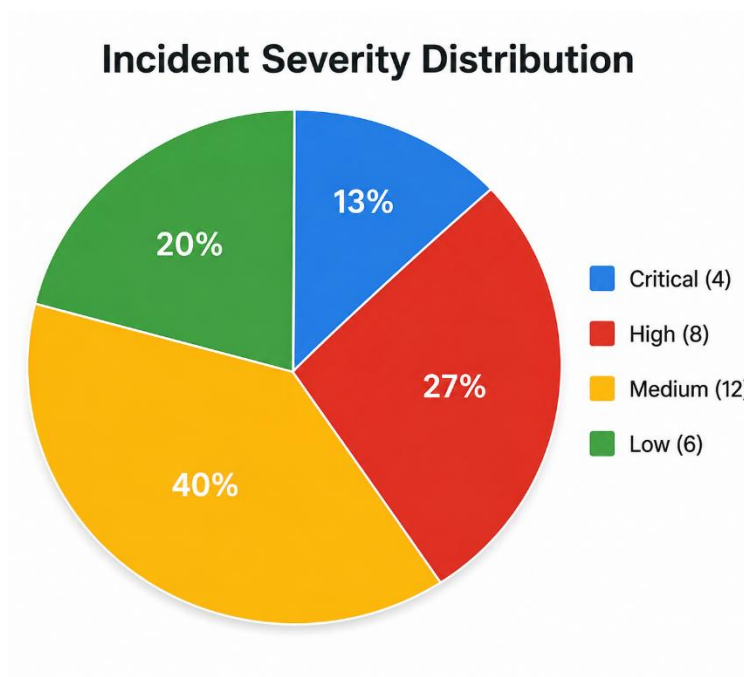


Figure 11: Incident Severity Distribution

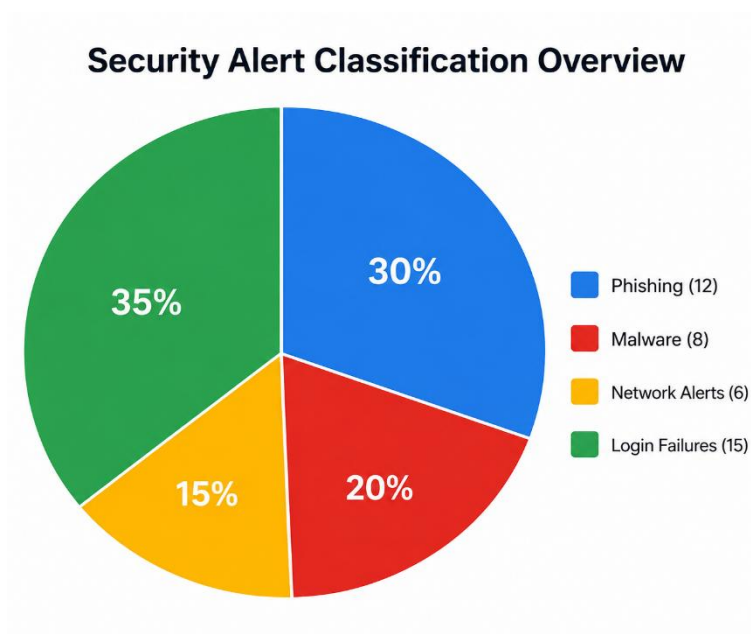


Figure 12: Security Alert Classification Overview

► Executive Summary:

The security metrics analysis demonstrated improved SOC monitoring efficiency and incident response performance during the simulated investigation activities. Multiple phishing alerts, malware detections, and suspicious login attempts were identified and investigated throughout the monitoring process. Security metrics such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and incident resolution rates were evaluated to measure operational effectiveness and identify opportunities for improvement.

The reporting activities highlighted the importance of continuous monitoring, automated alert handling, and proactive threat detection methodologies within modern SOC environments. Executive-level reporting procedures were also used to summarize investigation outcomes, operational metrics, and recommendations for improving security monitoring capabilities and incident response readiness.



► Observation:

The generated security metrics and charts provided visibility into incident trends, alert severity distribution, and SOC operational performance. The analysis demonstrated how reporting dashboards assist management teams in evaluating security monitoring effectiveness and response efficiency.

► Learning Outcome:

This activity improved understanding of SOC performance metrics, executive reporting methodologies, incident trend analysis, and dashboard visualization techniques commonly used during cybersecurity operations management.

SECTION 8 - CAPSTONE PROJECT

Capstone Project: Comprehensive SOC Incident Response Workflow

► Objective:

The objective of this capstone project was to simulate a complete SOC incident response lifecycle including attack simulation, threat detection, alert triage, evidence analysis, incident containment, reporting, and post-incident review procedures within a controlled monitoring environment.

The project was designed to integrate all concepts learned throughout the SOC training program into a practical incident response scenario. It also provided hands-on experience in coordinating multiple investigation activities while following standard SOC workflows and security best practices.



```
Session Actions Edit View Help

(kali@kali)~$ msfconsole
Metasploit tip: After running db_nmap, be sure to check out the result
of hosts and services

Unable to handle kernel NULL pointer dereference at virtual address 0xd34db33f
EFLAGS: 00010046
eax: 00000001 ebx: f77c8c00 ecx: 00000000 edx: f77f0001
esi: 003bfb14 edi: 00237f55 ebp: 00237f04 esp: 00237f08
ds: 0018  es: 0018  ss: 0018
Process Swapper (Pid: 0, process nr: 0, stackpage=00377000)

Stack: 90909090909090909090909090909090
90909090909090909090909090909090
90909090909090909090909090909090
90909090909090909090909090909090
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fffffffcfffffffcfffffffcfffffffc
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fffffffcfffffffcfffffffcfffffffc
fffffffcfffffffcfffffffcfffffffc

Code: 00 00 00 00 M3 T4 SP L0 IT FR 4M 3W OR K1 V3 R5 I0 N5 00 00 00 00
Alee, Killing Interrupt handler
ctrl + c will kill the idle task!
In swapper task - not syncing

==[ metasploit v6.4.126-dev ]
+ -- --[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]
+ -- --[ 432 post - 49 encoders - 14 nops - 12 evasion ]
```

Figure 13: Simulated Attack Environment using Metasploit

```
Session Actions Edit View Help

(kali@kali)~$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 0.0.0.0:15601 0.0.0.0:* LISTEN
tcp6 0 0 :::9200 :::* LISTEN
tcp6 0 0 127.0.0.1:19300 :::* LISTEN
tcp6 0 0 ::1:9200 :::* LISTEN
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49846 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:39324 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:39314 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49850 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:39310 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:50158 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:55552 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:40868 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:55544 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:40866 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:58132 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49442 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:58148 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:40878 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49422 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:40894 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:55568 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:55548 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49862 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:52124 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49438 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:49436 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:58146 TIME_WAIT
tcp6 0 0 127.0.0.1:19200 127.0.0.1:39326 TIME_WAIT
udp 0 0 192.168.112.128:68 192.168.112.254:67 ESTABLISHED
raw6 0 0 :::58 :::* 7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags Type State I-Node Path
unix 3 [ ] STREAM CONNECTED 11934
unix 3 [ ] STREAM CONNECTED 12351
unix 3 [ ] STREAM CONNECTED 10137 /run/user/1000/bus
unix 3 [ ] STREAM CONNECTED 9130
unix 3 [ ] STREAM CONNECTED 10078
unix 3 [ ] STREAM CONNECTED 10739
unix 3 [ ] DGRAM CONNECTED 10366
unix 2 [ ACC ] STREAM LISTENING 948 /run/pcscd/pcscd.comm
unix 3 [ ] STREAM CONNECTED 5960
unix 3 [ ] DGRAM CONNECTED 7645
unix 2 [ ACC ] STREAM LISTENING 949 /run/ssh-unix-local/socket
```

Figure 14: Network Connection Monitoring during Incident Investigation



```
Session Actions Edit View Help
kali@kali: ~
$ ps aux
USER      PID CPU % MEM % VSZ  RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 23932 14320 ?        Ss   02:10   0:01 /sbin/init splash
root         2  0.0  0.0      0   0 ?        S    02:10   0:00 [kthreadd]
root         3  0.0  0.0      0   0 ?        S    02:10   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-kvfree_rcu_reclaim]
root         5  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-rcu_gp]
root         6  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-sync_wq]
root         7  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-slab_flushwq]
root         8  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-netns]
root        12  0.0  0.0      0   0 ?        I    02:10   0:00 [kworker/u128:0-ipv6_addrconf]
root        13  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-mm_percpu_wq]
root        14  0.0  0.0      0   0 ?        I    02:10   0:00 [rcu_tasks_kthread]
root        15  0.0  0.0      0   0 ?        I    02:10   0:00 [rcu_tasks_rude_kthread]
root        16  0.0  0.0      0   0 ?        I    02:10   0:00 [rcu_tasks_trace_kthread]
root        17  0.0  0.0      0   0 ?        S    02:10   0:00 [ksoftirqd/0]
root        18  0.0  0.0      0   0 ?        I    02:10   0:02 [rcu_preempt]
root        19  0.0  0.0      0   0 ?        S    02:10   0:00 [rcu_exp_par_gp_kthread_worker/1]
root        20  0.0  0.0      0   0 ?        S    02:10   0:00 [rcu_exp_gp_kthread_worker]
root        21  0.0  0.0      0   0 ?        S    02:10   0:00 [migration/0]
root        22  0.0  0.0      0   0 ?        S    02:10   0:00 [idle_inject/0]
root        23  0.0  0.0      0   0 ?        S    02:10   0:00 [cpuhp/0]
root        24  0.0  0.0      0   0 ?        S    02:10   0:00 [cpuhp/1]
root        25  0.0  0.0      0   0 ?        S    02:10   0:00 [idle_inject/1]
root        26  0.0  0.0      0   0 ?        S    02:10   0:00 [migration/1]
root        27  0.0  0.0      0   0 ?        S    02:10   0:00 [ksoftirqd/1]
root        29  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/1:0H-kblockd]
root        30  0.0  0.0      0   0 ?        S    02:10   0:00 [cpuhp/2]
root        31  0.0  0.0      0   0 ?        S    02:10   0:00 [idle_inject/2]
root        32  0.0  0.0      0   0 ?        S    02:10   0:00 [migration/2]
root        33  0.0  0.0      0   0 ?        S    02:10   0:00 [ksoftirqd/2]
root        36  0.0  0.0      0   0 ?        S    02:10   0:00 [cpuhp/3]
root        37  0.0  0.0      0   0 ?        S    02:10   0:00 [idle_inject/3]
root        38  0.0  0.0      0   0 ?        S    02:10   0:00 [migration/3]
root        39  0.0  0.0      0   0 ?        S    02:10   0:00 [ksoftirqd/3]
root        41  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/3:0H-events_highpri]
root        48  0.0  0.0      0   0 ?        S    02:10   0:00 [kdevtmpfs]
root        49  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-inet_frag_wq]
root        50  0.0  0.0      0   0 ?        S    02:10   0:00 [ksuidtd]
root        51  0.0  0.0      0   0 ?        S    02:10   0:00 [khungtaskd]
root        52  0.0  0.0      0   0 ?        S    02:10   0:00 [oom_reaper]
root        55  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-writeback]
root        56  0.0  0.0      0   0 ?        S    02:10   0:01 [kcompactd0]
root        57  0.0  0.0      0   0 ?        SN   02:10   0:00 [ksmd]
root        58  0.0  0.0      0   0 ?        SN   02:10   0:01 [khugepaged]
root        59  0.0  0.0      0   0 ?        Ic   02:10   0:00 [kworker/R-kintegrityd]
```

Figure 15: Process Monitoring during SOC Investigation

```
Session Actions Edit View Help
kali@kali: ~
$ echo "test" > test.txt
kali@kali: ~
$ sha256sum test.txt
f2ca1bb6c7e987d86dafe4687e579fce76b37e4e93b7605822da52e6ccc26fd2 test.txt
kali@kali: ~
$
```

Figure 16: Evidence Integrity Verification using SHA256 Hashing



INCIDENT TIMELINE TABLE:

Timestamp	Activity	Status
10:00 AM	Attack Simulation Initiated	Completed
10:05 AM	Suspicious Connection Detected	Detected
10:10 AM	Alert Generated	Completed
10:15 AM	SOC Investigation Started	Completed
10:20 AM	Evidence Collected	Completed

INCIDENT ANALYSIS TABLE:

Source IP	Alert Description	Severity	MITRE Technique
192.168.1.102	Phishing Simulation	High	T1566
192.168.1.105	Exploitation Attempt	Critical	T1210
192.168.1.110	Command Execution Activity	Medium	T1059

► Executive Summary:

The capstone project simulated a complete SOC investigation workflow involving attack simulation, monitoring, alert generation, evidence analysis, and incident response activities. Multiple system monitoring techniques were used to identify suspicious network connections, active processes, and potential attacker behavior during the investigation process. Simulated attack activities aligned with the MITRE ATT&CK framework were analyzed to evaluate SOC monitoring effectiveness and incident response readiness.



The investigation workflow included alert triage, evidence validation, process analysis, hash verification, and root cause analysis procedures commonly followed during enterprise SOC operations. The collected evidence and monitoring outputs were reviewed to improve detection visibility and enhance understanding of end-to-end cybersecurity incident handling methodologies.

► **Explanation:**

The capstone project integrated multiple SOC operational procedures into a single incident response simulation workflow. Attack simulation activities were performed using the Metasploit Framework, while monitoring commands were used to identify suspicious activity and validate generated alerts. Evidence collection, hash verification, and process analysis activities were also performed to support the investigation process and improve incident response coordination.

► **Observation:**

The capstone investigation successfully demonstrated how SOC teams identify, analyze, contain, and document cybersecurity incidents within monitored environments. The investigation workflow highlighted the importance of continuous monitoring, automated alert handling, evidence preservation, and structured incident response coordination during security operations.

► **Learning Outcome:**

This capstone project improved understanding of complete SOC operational workflows including adversary emulation, threat hunting, alert triage, evidence analysis, incident containment, executive reporting, and continuous security improvement methodologies.



Final Observation:

The Week 4 SOC activities successfully demonstrated advanced cybersecurity monitoring, threat hunting, incident response automation, evidence analysis, adversary emulation, and executive reporting procedures within a simulated SOC environment. Multiple practical activities including attack simulation, alert triage, root cause analysis, and evidence validation were performed to improve understanding of modern SOC workflows and cybersecurity operations management. The overall investigation process highlighted the importance of continuous monitoring, automated response workflows, structured incident handling, and proactive threat detection methodologies during cybersecurity operations. The capstone project further demonstrated how multiple SOC processes integrate together to improve incident response effectiveness and organizational security readiness.

Conclusion:

The Week 4 practical activities provided valuable understanding of advanced SOC operational procedures including threat hunting, alert triage, evidence preservation, adversary emulation, security metrics analysis, and incident response automation. Multiple practical investigations and simulated attack scenarios were analyzed to improve knowledge of cybersecurity monitoring workflows and defensive security operations.

The capstone project successfully demonstrated a complete SOC investigation lifecycle involving attack simulation, detection, triage, containment, reporting, and post-incident analysis procedures. The activities also improved understanding of MITRE ATT&CK mapping, root cause analysis methodologies, executive reporting techniques, and continuous security improvement practices commonly followed within enterprise SOC environments.

Overall, the task enhanced practical SOC investigation skills, incident response coordination techniques, and cybersecurity operational awareness required for modern security monitoring and defense operations.